Lab：Finding the minimum value through ultrasound

This experiment explores the measurement method of ultrasonic sensors, using a timer or button to control the car to rotate once and find the direction with the maximum distance to move forward 20cm.

1 Ultrasonic module

The HC-SR04 ultrasonic module can provide distance sensing function between 2cm and 400cm, and the measurement accuracy can reach 3mm. The module includes an ultrasonic transmitter, receiver, and control circuit.



Four interface terminals:

Vcc pin for 5V power supply, GND pin for ground wire, Trig pin for control signal, Echo pin is signal output.

Electrical parameters：

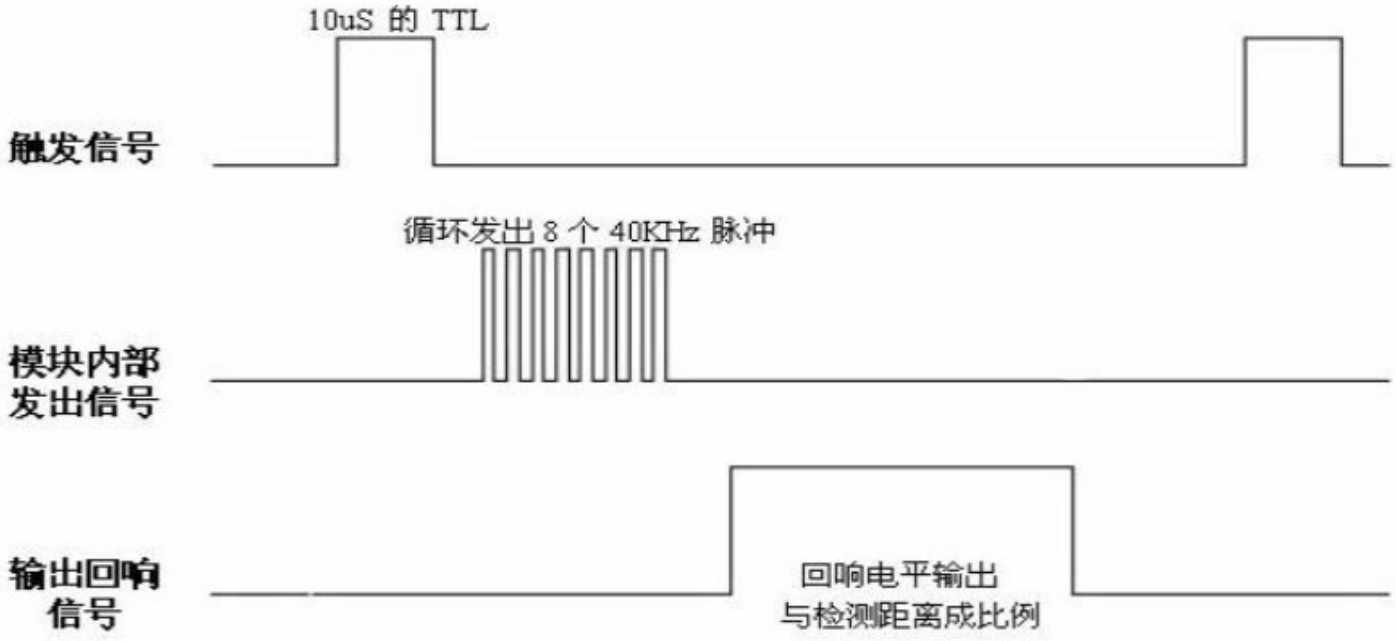
电气参数	HC-SR04超声波模块
工作电压	DC 5 V
工作电流	15mA
工作频率	40kHz
最远射程	4m
最近射程	2cm
测量角度	15 度
输入触发信号	10uS 的 TTL 脉冲
输出回响信号	输出 TTL 电平信号，与射程成比例
规格尺寸	45*20*15mm

Operational principle：

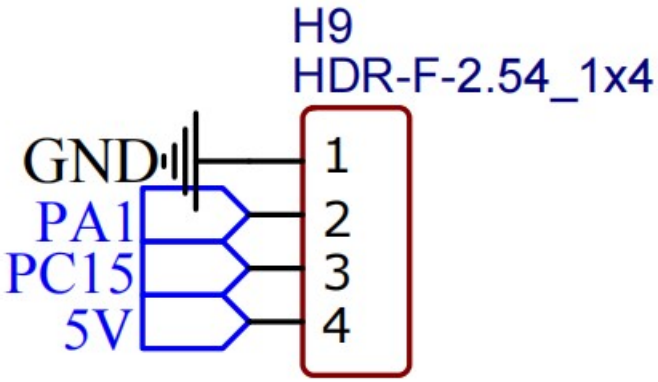
The IO port is used to trigger the measurement program for Trig, providing a high-level signal at least 10us. The module automatically sends 8 40kHz square waves and detects whether there is a signal return; There is a signal return, and a high level is output through the IO port Echo. The duration of the high level is the time from the emission to the return of the ultrasonic wave. Test distance=(high-level time * sound speed (340M/S))/2.

If a control port sends a high level of 10us or more, it can wait for the high level output at the receiving port. As soon as there is an output, the timer can be turned on for timing. When this port becomes low, the value of the timer can be read, which is the time for this distance measurement, and then the distance can be calculated. Such continuous cyclic ranging can achieve the value of mobile measurement distance.

Time sequence diagram:



In the schematic diagram of the C10B development board, the interface of the ultrasonic ranging module is shown in the following figure.



2 Test program

The serial port display reference program for ultrasonic measurement is attached.

2.1 Experimental requirements

Firstly, students should test the serial port to familiarize themselves with the performance of the ultrasonic ranging module. They can use their hands or obstacles to block the test, and place a screenshot of the student card to save.

Secondly, students use a timer or set a button, and after pressing it, the car moves more than one circle around the center of the circle in the designated environment. The car uses an ultrasonic ranging module while moving, and filtering algorithms are required to maintain accuracy in ultrasonic ranging. Students should test the distance value of the car as much as possible for one revolution, and set an array or other methods in the program to find the maximum distance value around the car. Then, the car will automatically rotate to the direction of the maximum distance and move forward 10-15cm. The more precise the maximum direction, the higher the score.

Finally, after finding the maximum direction, the car moves forward and stops 1cm in front of the set sign using auxiliary assistance such as infrared distance sensors or rotary servos. And shoot a video to submit the report.